

Roberto André Mossi Milla

Computer Vision Engineer | Machine Learning | AI Systems

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Summary

Computer Vision and Machine Learning Engineer building production AI systems spanning computer vision, multimodal AI, OCR, and full-stack deployment. Experience includes YOLOv8 detection and tracking, multimodal model integration, GPU-accelerated AI research, image processing, model retraining workflows, and AI systems that reduce previously manual processes from days or weeks to minutes.

Experience

Ingeniería Digital

May 2026 – Present

Software Engineer (AI Systems)

Honduras

- Built a multimodal AI vehicle-damage assessment platform that reduced assessment workflows from more than one week to approximately 1–2 minutes, automatically generating damage analyses and insurance estimates from vehicle imagery and VIN information.
- Developed the end-to-end application across React, Next.js, TypeScript, PostgreSQL, AWS S3, and AI services, integrating AWS Rekognition for vehicle localization, Gemini Flash for damage detection, VIN-history retrieval, and automated insurance-estimation workflows.
- Designed an interactive AI evaluation workflow using Konva that allows users to inspect, move, and correct detection regions, persist annotations, and evaluate different models, processing layers, and AI combinations.
- Built Playwright-based data-acquisition services to retrieve and normalize vehicle information and VIN history from external sources for valuation and insurability analysis.
- Built an AI-assisted medical document generation platform from scratch, combining Amazon Textract OCR, OpenAI APIs, structured document generation, and cloud storage; reduced report preparation from approximately one week to 1 day, with final document generation completed in under one minute.
- Architected secure full-stack document workflows using React, Next.js, TypeScript, PostgreSQL, Auth0, and AWS S3 for document upload, OCR processing, AI-assisted editing, versioning, and management.

Gannon University

Aug. 2024 – May 2025

Computer Vision Engineer — Industry-Sponsored Project

Erie, PA

- Engineered a real-time industrial computer-vision pipeline using LiDAR-triggered image capture and Ultralytics YOLOv8 to identify packages and pallets entering a warehouse and automatically record qualifying detection events for downstream damage analysis.
- Developed a live monitoring dashboard displaying YOLOv8 detection bounding boxes and events, with administrative tools allowing operators to move and correct annotations and submit revised data for model retraining.
- Retrained YOLOv8 models using curated datasets and improved image quality through contrast, color, filtering, spatial partitioning, and image-reconstruction techniques to support more reliable inference.
- Containerized distributed inference and data-processing services using Docker and integrated C++, Python, and Rust components for real-time system communication.
- Helped migrate the inference system from Raspberry Pi hardware after thermal and performance limitations were reached, supporting deployment on NVIDIA Jetson hardware while maintaining near-real-time visual feedback.
- Took primary ownership of the monitoring and visualization interface, integrating computer-vision events with TimescaleDB and backend services for operational review.

Gannon University

Aug. 2024 – May 2025

Research Assistant — Evolutionary Artificial Intelligence

Erie, PA

- Conducted undergraduate research on autonomous multi-agent learning and evolutionary computation, investigating emergent behavior in dynamic simulated environments using NeuroEvolution of Augmenting Topologies (NEAT).
- Independently implemented NEAT in C++ and designed a numerical genome representation encoding agent sensing, actions, and neural-network structure for evolutionary optimization across generations.
- Developed a procedurally generated 3D Unreal Engine environment capable of producing randomized or seeded halls, floors, walls, and objects for repeated multi-agent experiments using C++ and CUDA.
- Contributed to the peer-reviewed paper “Tag AI-Sandbox,” published through the American Society for Engineering Education (ASEE) in 2025 and presented at a Sigma Xi conference at Penn State University.

Dinant

Jun. 2023 – Jul. 2023

Full-Stack Developer

Honduras

- Led end-to-end development of two production platforms used by two companies, replacing manual operational processes that previously required weeks with automated workflows completed in minutes; systems remain in use.
- Designed relational database schemas and full-stack workflows for contract management, digital signatures, automated PDF generation, repair tracking, machine maintenance, inventory deduction, and operational reporting.
- Collaborated directly with international stakeholders, gathering requirements, presenting iterative demonstrations, incorporating production feedback, and deploying finalized systems.

Open Source

Copilot-Lualine

2024 – Present

Creator & Maintainer

Open Source

- Created and maintain an open-source Neovim plugin providing interactive GitHub Copilot status indicators and controls for enabling, disabling, authentication, and logging; received 117+ GitHub stars and contributions from four community developers.

Technical Skills

Computer Vision & ML: Ultralytics YOLOv8, AWS Rekognition, Gemini Flash, OpenAI APIs, CUDA, Computer Vision, Multimodal AI, OCR, Image Analysis, Object Detection, Object Tracking, NEAT, Evolutionary Computation

Languages: Python, C++, Rust, TypeScript, JavaScript, SQL, Java, Bash, Lua

Systems & Data: Docker, Linux, TimescaleDB, PostgreSQL, Redis, REST APIs, Event-Driven Services, AWS S3, Amazon Textract

Application Engineering: React, Next.js, Node.js, Playwright, Konva, Auth0, GitHub Actions, CI/CD

Education & Recognition

Gannon University

2025

B.S. Computer Systems Engineering

Erie, PA

- Peer-reviewed publication with the American Society for Engineering Education (ASEE); research presentation at Sigma Xi.